



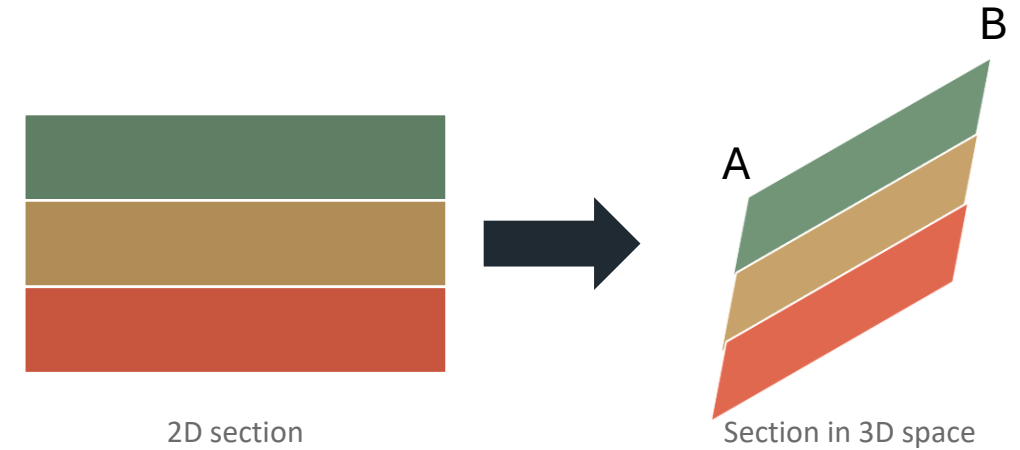
Geological 3D Cross Section

A QGIS plugin for geologists, written together with an artificial intelligence

Niccolò Iandelli · with Claude (Anthropic)

The problem to solve

- A geological cross section starts as a 2D drawing: distance along the trace on the horizontal axis, elevation on the vertical axis.
- To view it in the 3D model together with the boreholes, it must sit in its real geographic position, standing vertically along the section trace.



Parameters: section layer, horizontal axis, Point A, Point B, tilt angle, vertical exaggeration (x1–x10 or custom), Z offset.
Integrated in the Processing Toolbox and available from a toolbar button.

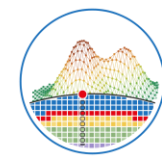
Goal

Give two real points (A on the left, B on the right) and get the section placed in position, optionally tilted, with vertical exaggeration and an elevation offset when exaggerated.

The workflow

1. Place the continuous-core boreholes with the Geoscience plugin (<https://plugins.qgis.org/plugins/geoscience/>)
2. Interpret the section generated with Geoscience (<https://plugins.qgis.org/plugins/geoscience/>)
3. Rotate the section into 3D with Geological 3D Cross Section
4. Style the section
5. View the section in 3D with QGIS THREEJS or with VISTA 3D

You need the GEOSCIENCE plugin



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Geoscience

Plugin ID: 1542 

Tools for Geoscience & Exploration

★★★★★ (227) votes

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About

Details

Versions

Geoscience: **Tools for Geoscience and Exploration**

- Drill hole display
- Drill hole sections
- Convert alpha/beta structure data
- Vector tools
- Raster tools
- Local grids

 Send ideas & requests to roleyhill@gmail.com

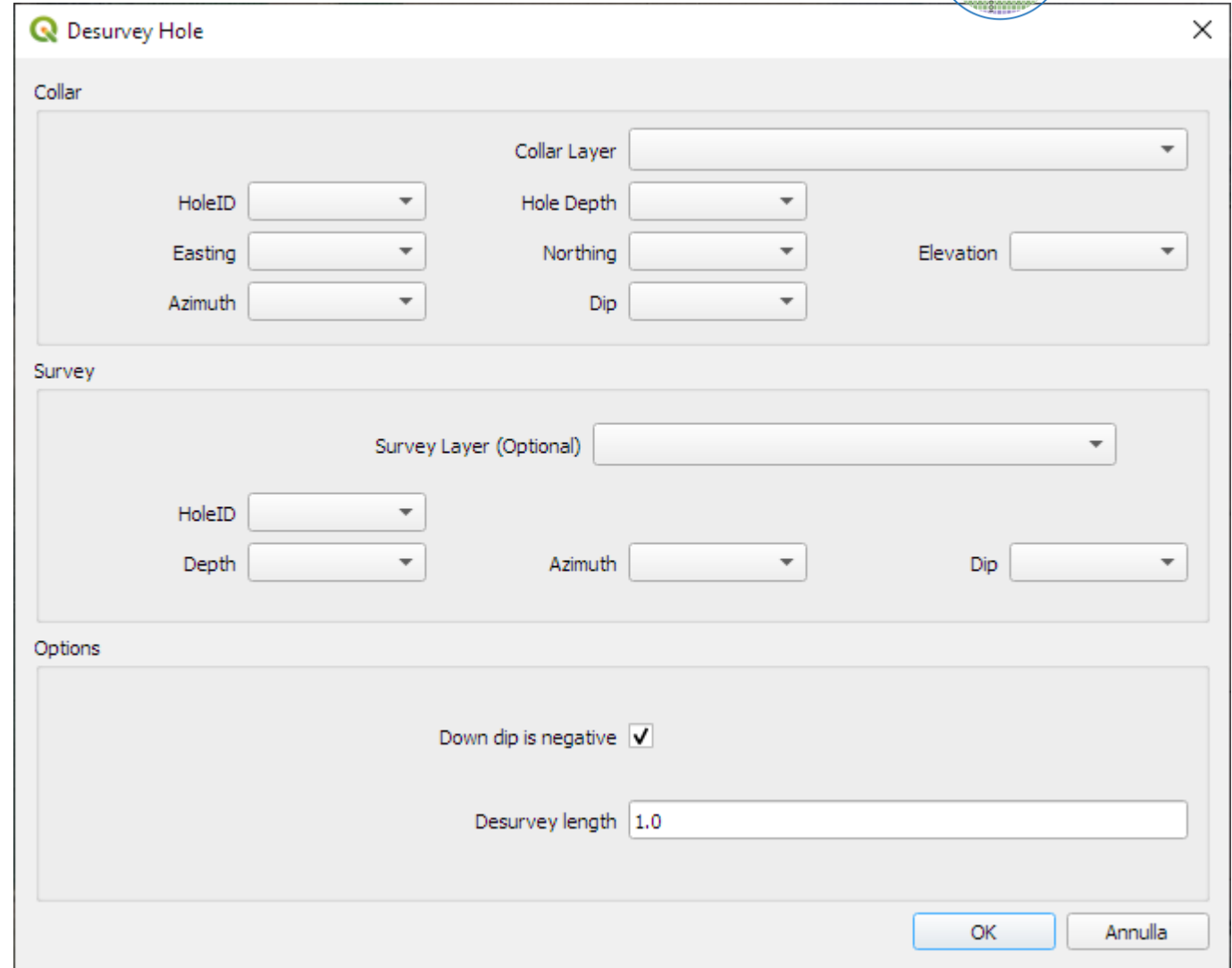
Set up Geoscience 1/4

Collar -> CSV

HoleID
Hole_Depth
Easting
Northing
Elevation
Azimuth (0)
Dip (-90)

Survey -> CSV

HoleID
Depth
Azimuth (0)
Dip (-90)



The screenshot shows the 'Desurvey Hole' software window. It is divided into three main sections: 'Collar', 'Survey', and 'Options'. The 'Collar' section contains dropdown menus for HoleID, Easting, Azimuth, Collar Layer, Hole Depth, Northing, Dip, and Elevation. The 'Survey' section contains a dropdown for Survey Layer (Optional), and dropdowns for HoleID, Depth, Azimuth, and Dip. The 'Options' section has a checkbox for 'Down dip is negative' which is checked, and a text input for 'Desurvey length' set to 1.0. At the bottom right are 'OK' and 'Annulla' buttons.

Desurvey Hole

Collar

Collar Layer

HoleID

Hole Depth

Easting

Northing

Elevation

Azimuth

Dip

Survey

Survey Layer (Optional)

HoleID

Depth

Azimuth

Dip

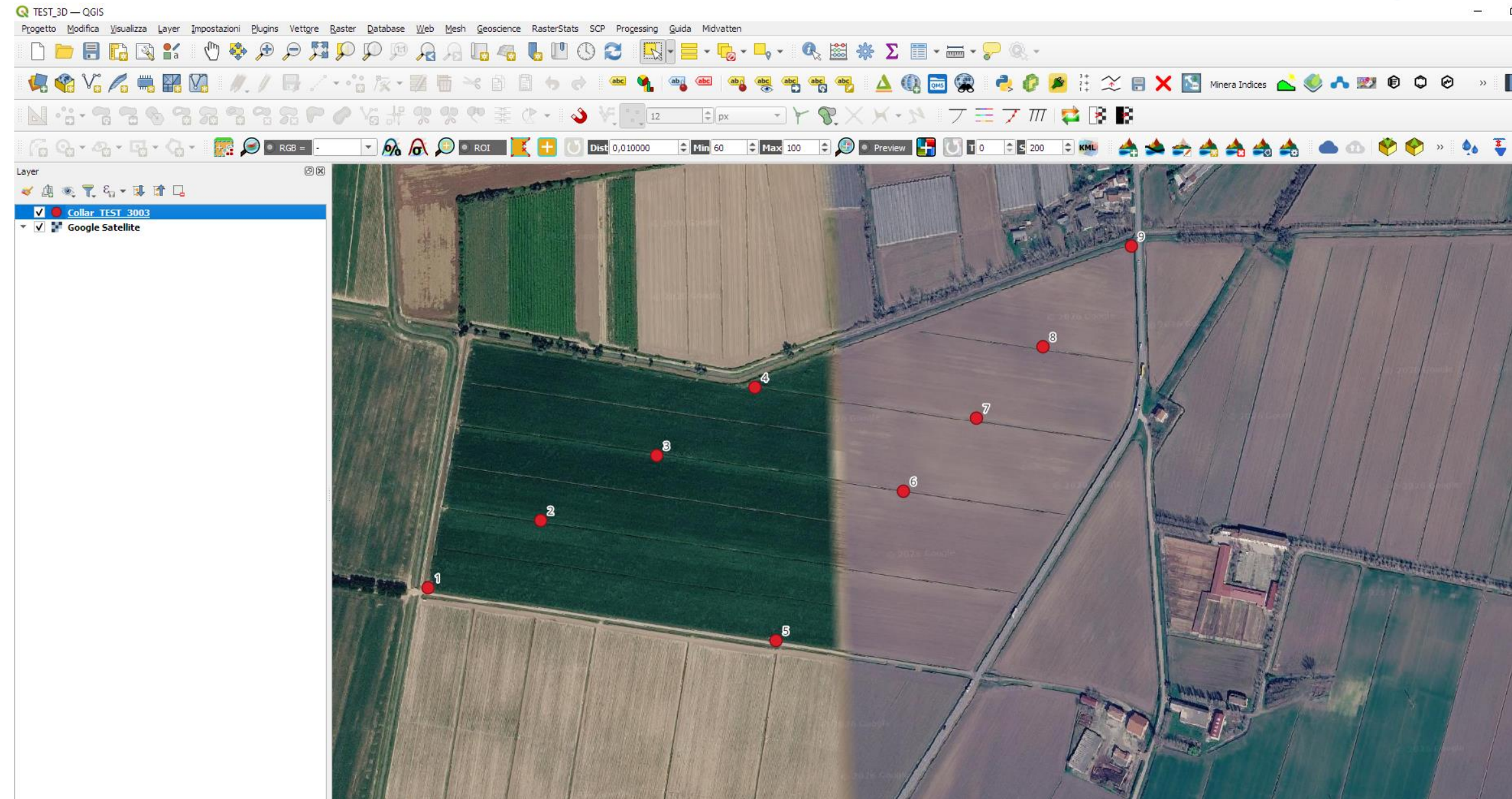
Options

Down dip is negative ☒

Desurvey length 1.0

OK Annulla

Set up Geoscience – COLLAR/SURVEY 2/4



Set up Geoscience – INTERVAL 3/4

Salvataggio automatico ☐ Interval_TEST_3003.csv

File Home Inserisci Layout di pagina Formule Dati Revisione Visualizza Automatizza Sviluppo Guida

Appunti Carattere Allineamento Numeri

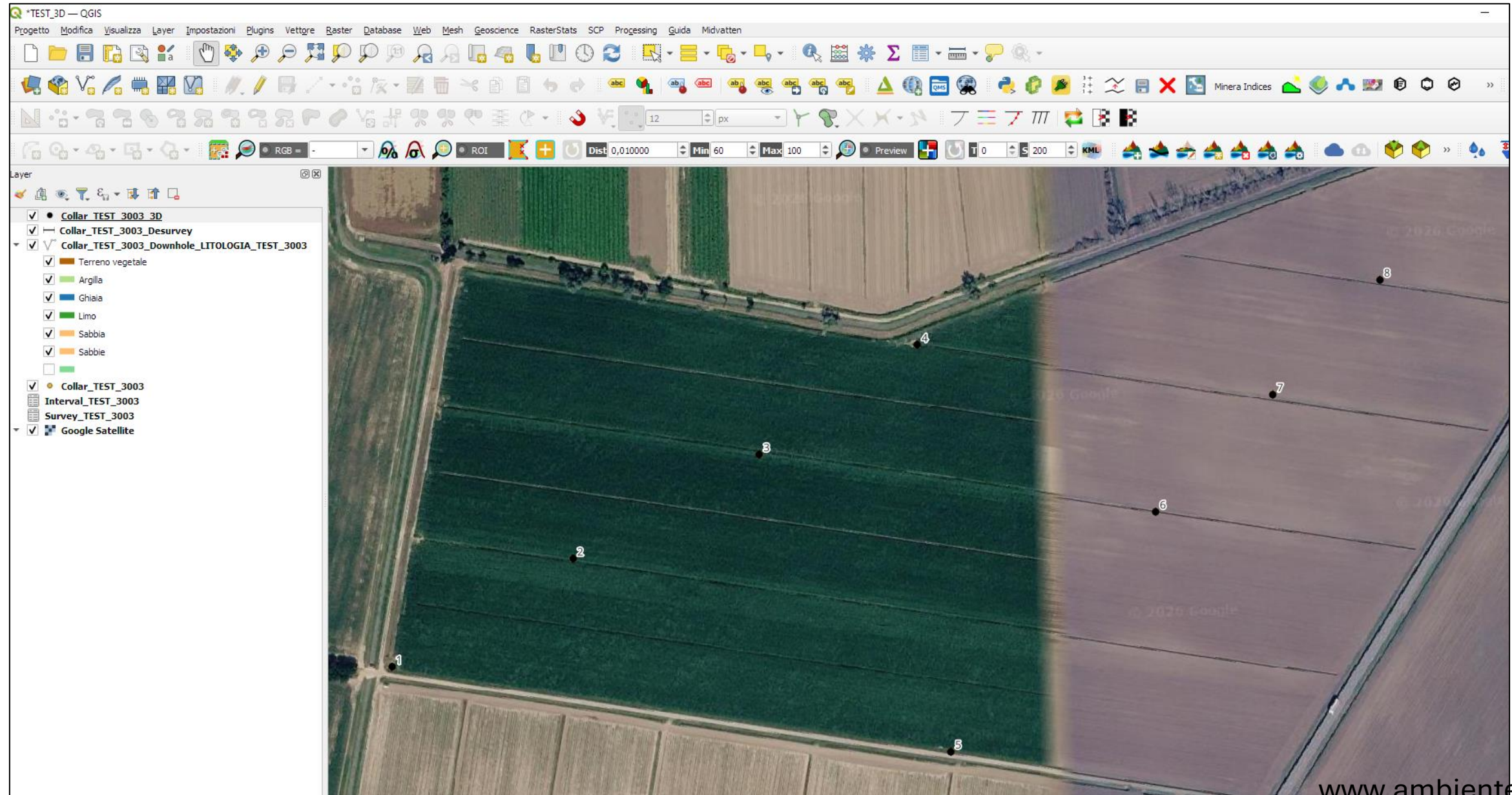
K15

	A	B	C	D	E	F	G	H	I	J	K
1	ID	HOLE_ID	SONDAGGIO	FROM	TO	DESCR	GROUP				
2	1	1	S1	0	0,7	Terreno agrario limoso argilloso	Terreno vegetale				
3	2	1	S1	0,7	3	Limo marrone debolmente sabbioso	Limo				
4	3	1	S1	3	5	Limo e argilla	Limo				
5	4	1	S1	5	7	Argille grigie	Argilla				
6	5	1	S1	7	12	Sabbie limose	Sabbia				
7	6	1	S1	12	15	Ghiaia in matrice sabbiosa	Ghiaia				
8	7	2	S2	0	1	Terreno agrario limoso argilloso	Terreno vegetale				
9	8	2	S2	1	3	Limo marrone debolmente sabbioso	Limo				
10	9	2	S2	3	5	Limo e argilla	Limo				
11	10	2	S2	5	9	Argille grigie	Argilla				
12	11	2	S2	9	12,5	Sabbie limose	Sabbia				
13	12	2	S2	12,5	20	Ghiaia in matrice sabbiosa	Ghiaia				
14	13	3	S3	0	1	Terreno agrario limoso argilloso	Terreno vegetale				
15	14	3	S3	1	5	Limo e argilla	Limo				
16	15	3	S3	5	12	Sabbie limose	Sabbie				
17	16	3	S3	12	15	Argille grigie	Argilla				
18	17	4	S4	0	1	Terreno agrario limoso argilloso	Terreno vegetale				
19	18	4	S4	1	5	Limo e argilla	Limo				
20	19	4	S4	5	8	Sabbie limose	Sabbia				
21	20	4	S4	8	10	Argille grigie	Argilla				
22	21	4	S4	10	15	Ghiaia in matrice sabbiosa	Ghiaia				
23	22	5	S5	0	0,8	Terreno agrario limoso argilloso	Terreno vegetale				
24	23	5	S5	0,8	4	Limo marrone debolmente sabbioso	Limo				

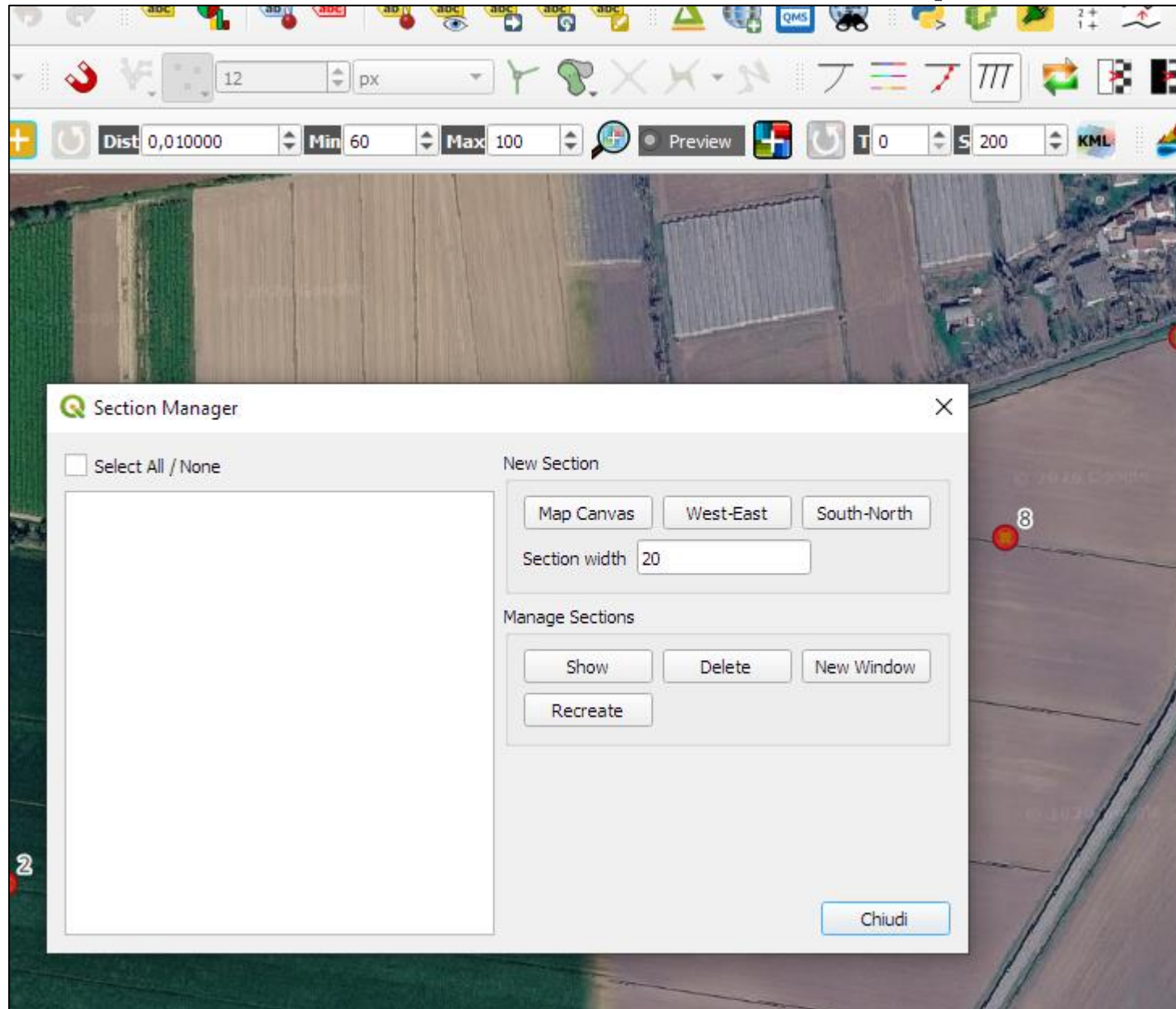
Interval -> CSV

HoleID
SONDAGGIO
FROM
TO
DESCR
GROUP

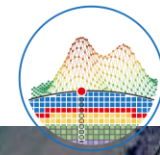
Set up Geoscience 4/4



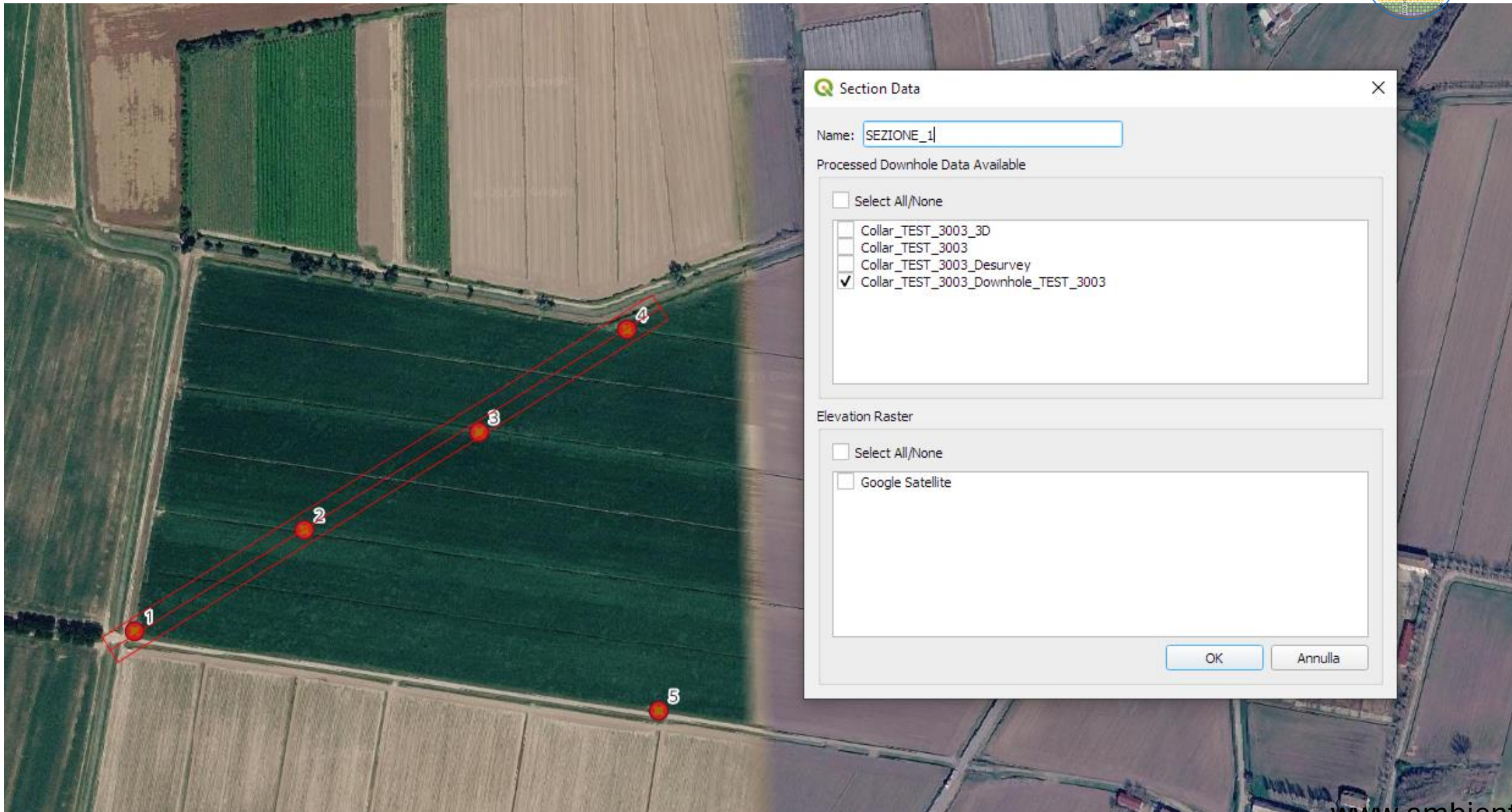
Build the section with Geoscience 1/3



Build the section with Geoscience 2/3



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Section Data

Name:

Processed Downhole Data Available

☐ Select All/None

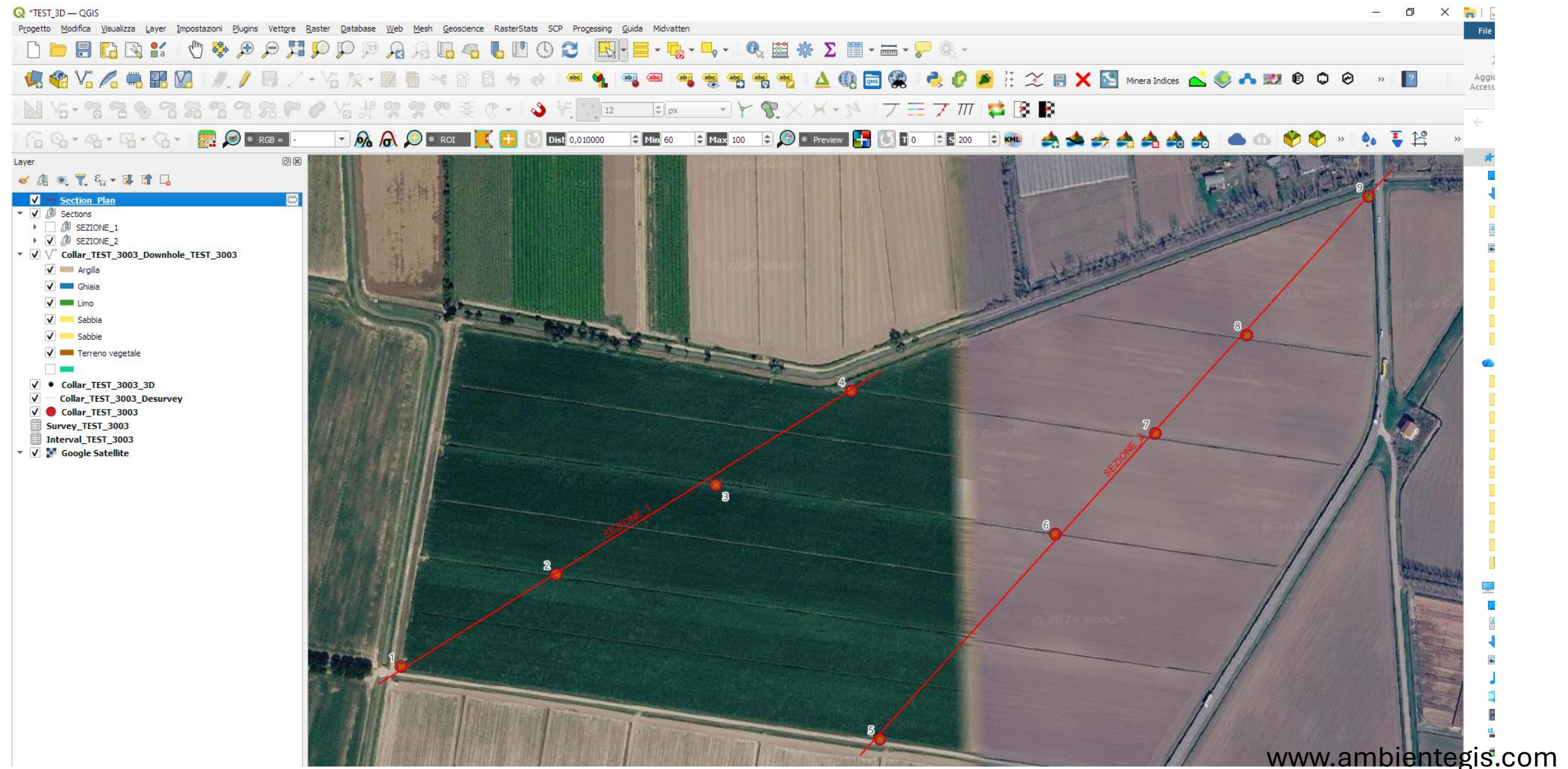
- ☐ Collar_TEST_3003_3D
- ☐ Collar_TEST_3003
- ☐ Collar_TEST_3003_Desurvey
- ☒ Collar_TEST_3003_Downhole_TEST_3003

Elevation Raster

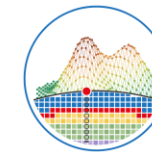
☐ Select All/None

☐ Google Satellite

Build the section with Geoscience 3/3



INTERPRET THE SECTION 1/2



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QGIS *TEST_3D — QGIS

Progetto Modifica Visualizza Layer Impostazioni Plugins Vettore Raster Database Web Mesh Geoscience RasterStats SCP Processing Guida Midvatten

Layer

- Section_Plan
 - Sections
 - SEZIONE_1
 - S_SEZIONE_1_3003
 - Terreno vegetale
 - Argilla
 - Ghiaia
 - Limo
 - Sabbia
 - Sabbie
 - SEZIONE_2
 - S_SEZIONE_2_3003
 - Terreno vegetale
 - Argilla
 - Ghiaia
 - Limo
 - Sabbia
 - Sabbie
- Decorations
 - Collar_TEST_3003_3D
 - Collar_TEST_3003_Desurvey
 - Collar_TEST_3003_Downhole_LITOLOGIA_TEST_3003
 - Terreno vegetale
 - Argilla
 - Ghiaia
 - Limo
 - Sabbia
 - Sabbie

Layer

- Section_Plan
 - Sections
 - SEZIONE_1
 - SEZIONE_1_3003
 - SEZIONE_2
 - SEZIONE_2_3003
- Collar TEST 3003 Downhole TEST 3003

RGB = -

ROI

Dist: 0,010000

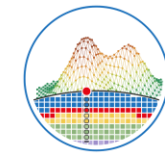
Min: 60 Max: 100

Preview

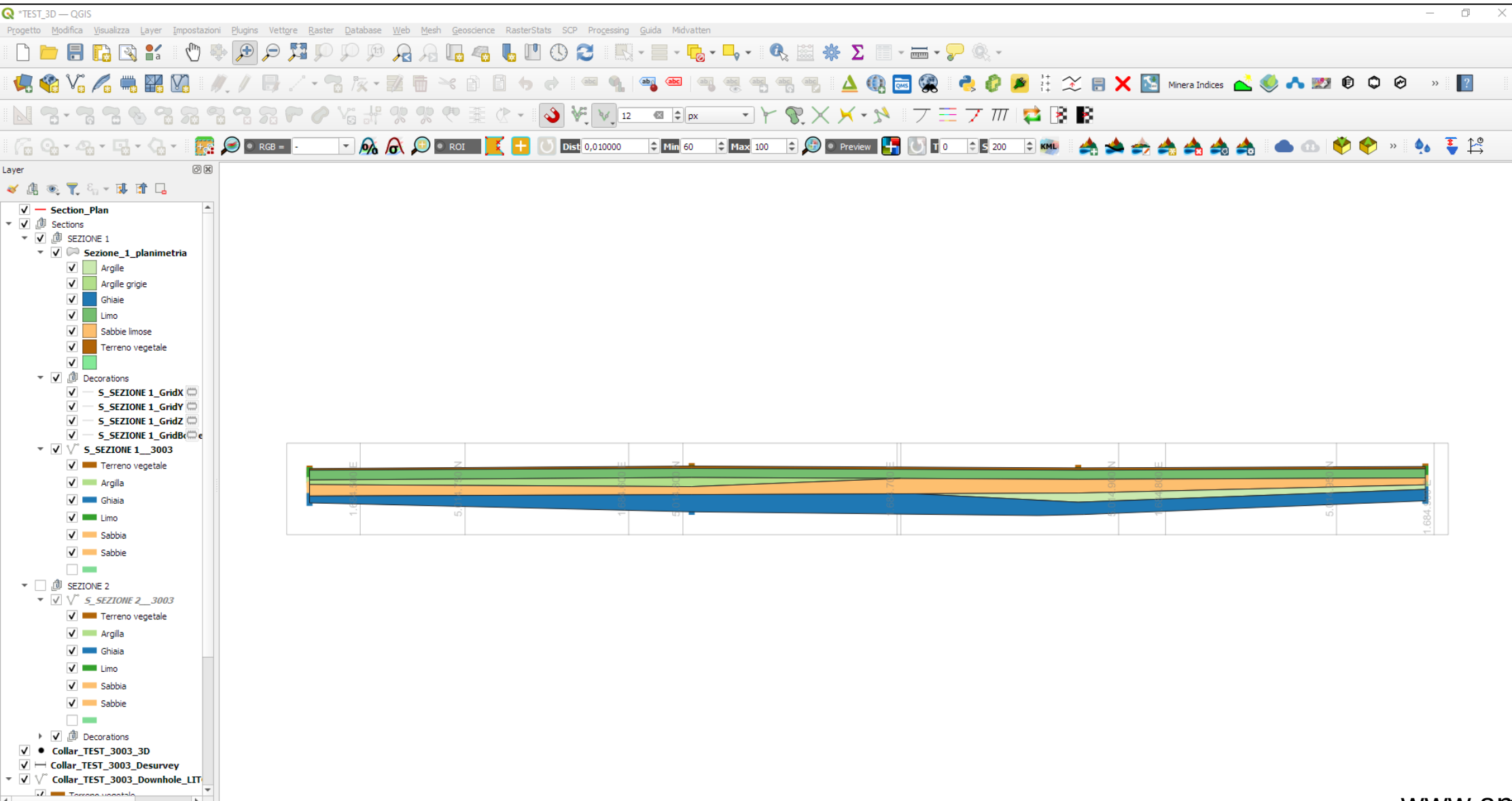
T: 0 S: 200 KML

5.014.700 N	1.685.000 E	5.014.800 N	1.685.100 E	5.014.900 N	1.685.200 E	5.015.000 N	1.685.300 E	5.015.100 N	1.685.400 E
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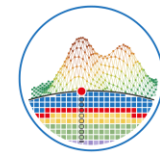
INTERPRET THE SECTION 2/2



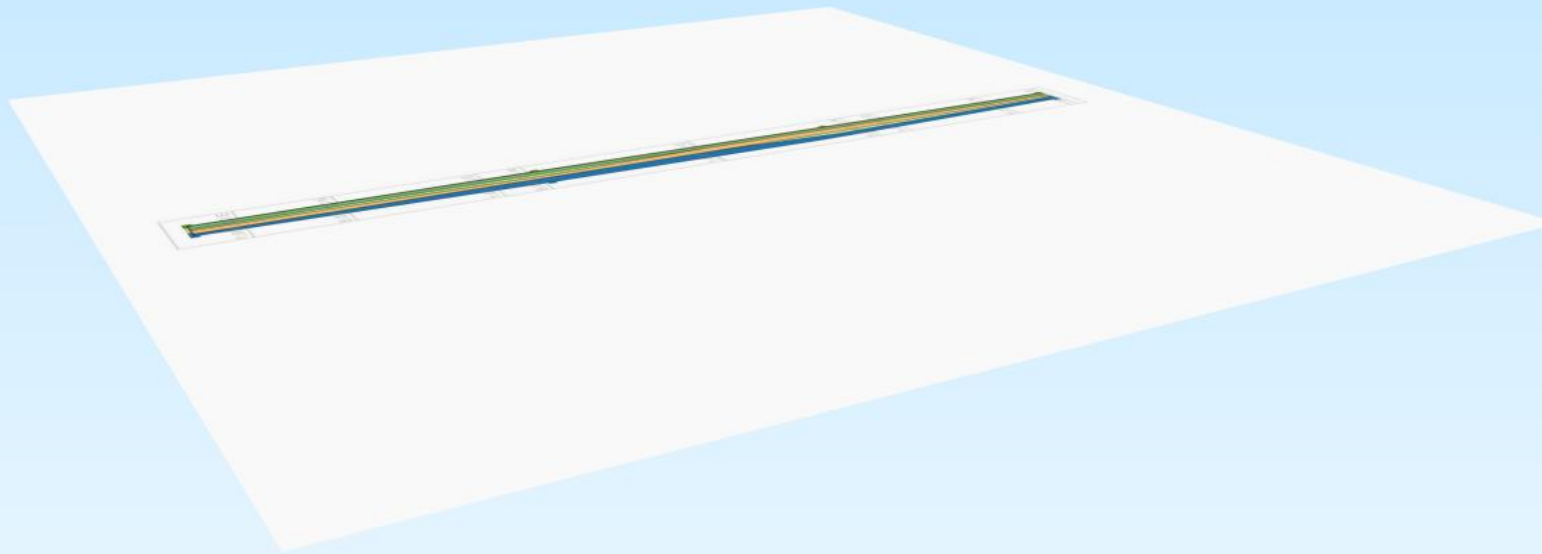
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View the section in 2D

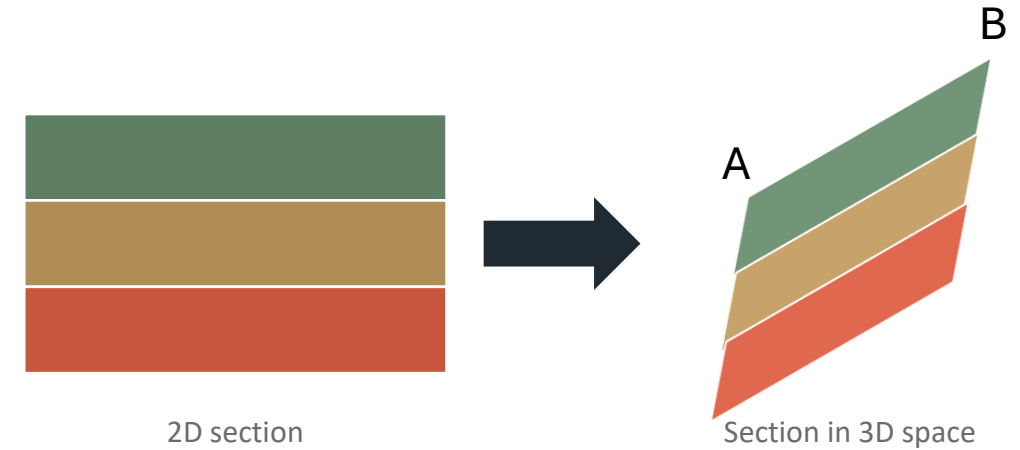


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The problem to solve

- A geological cross section starts as a 2D drawing: distance along the trace on the horizontal axis, elevation on the vertical axis.
- To view it in the 3D model together with the boreholes, it must sit in its real geographic position, standing vertically along the section trace.

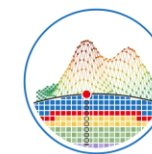


Parameters: section layer, horizontal axis, Point A, Point B, tilt angle, vertical exaggeration (x1–x10 or custom), Z offset.
Integrated in the Processing Toolbox and available from a toolbar button.

Goal

Give two real points (A on the left, B on the right) and get the section placed in position, optionally tilted, with vertical exaggeration and an elevation offset when exaggerated.

Download and install Geological 3D Cross Section



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Plugin | Tutto (3589)

Tutto

Installati

Non installati

Aggiornabile

Nuovo

Installa da ZIP

Impostazioni

Geological

ArchToolkit

CRAG

Dip-Strike Tools

FieldMove Project Importer

FraXtex

GeoCongo AI

geol_qmaps

Geological 3D Cross Section

Geology from points and lines

GeoProfile

GeoTrace2

Geovita GIS Processing provider

Limiti Geo

LoopStructural

mappy

Midvatten

Mineral Exploration Web Services

Parallel Folds & Structural Tool

qgSurf

☒ qProf

Sec Interp

Shp2SSAP

Siligites

Structural Families Mapper

ThreePointMethod

THYRSIS

Weights of Evidence (WofE) Model

Plugin nuovo

Geological 3D Cross Section



Place a 2D geological cross-section into 3D space between two real-world points (A and B) / Posiziona una sezione geologica 2D nello spazio 3D tra due punti reali (A e B).

EN: Geological 3D Cross Section places a 2D geological cross-section (polygon layer, drawn on the XY plane) into real-world 3D space. Pick point A (left end) and point B (right end) on the map: azimuth and horizontal scale are computed automatically. Then tilt the section around the A-B line, apply a fixed vertical exaggeration (x2, x2.5, x5, x10 or custom) and a Z offset. The section can come from the Geoscience plugin or be digitised from scanned raster sections; in both cases it must be geometrically correct and to scale. The output (PolygonZ) can be viewed with the QGIS 3D Map View or the Qgis2threejs plugin. Further developments are already in progress. --- IT: Geological 3D Cross Section posiziona una sezione geologica 2D (poligoni disegnati sul piano XY) nello spazio 3D reale. Scegli sulla mappa il punto A (estremo sinistro) e il punto B (estremo destro): azimuth e scala orizzontale sono calcolati automaticamente. Puoi poi inclinare la sezione attorno alla linea A-B, applicare un'esagerazione verticale fissa (x2, x2.5, x5, x10 o personalizzata) e un offset di quota Z. La sezione può derivare dal plugin Geoscience oppure essere disegnata a partire da sezioni scannerizzate in raster; in entrambi i casi deve essere geometricamente corretta e in scala. Il risultato (PolygonZ) si visualizza con la Vista Mappa 3D di QGIS o con il plugin Qgis2threejs. Altri sviluppi sono già in lavorazione.

☆☆☆☆☆ 0 voto(i), 36 download

Etichette [geology](#), [cross section](#), [3d](#), [rotation](#), [geologia](#), [sezione](#), [geoscience](#)

Maggiori informazioni [homepage](#) [bug tracker](#) [repository del codice](#)

Autore Niccolò Iandelli (with the support of Claude)

Versione disponibile (stabile) [1.0](#) aggiornato a 30/09/2026 20:24 ora legale Europa occidentale

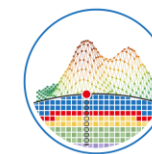
Aggiorna Tutto

Installa Plugin

Chiudi

Aiuto

Set the parameters



Geologia - Ruota sezione 2D in 3D (punti A/B)

Parametri Log

Sezione 2D (poligoni)
Sezione_1_planimetria [EPSG:3003] A

Asse orizzontale della sezione nel layer originale
X esprime la distanza lungo la sezione (Y è la quota) B

Punto A (estremo sinistro della sezione)
C

Punto B (estremo destro della sezione)
D

Angolo di rotazione verticale attorno alla linea A-B (gradi, 0 = sezione verticale, 90 = sezione sdraiata)
0,000000 E

Esagerazione verticale (applicata alla quota prima della rotazione)
x1 (nessuna) F

Valore esagerazione personalizzata (usato solo se sopra è selezionato 'Personalizzata...') [opzionale]
1,000000 G

Offset di quota Z (aggiunto dopo la rotazione)
0,000000 H

0%

Avanzato Esegui come Processo in Serie...

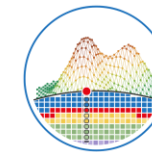
Esegui Annulla Chiudi

Ruota sezione 2D in 3D (punti A/B)

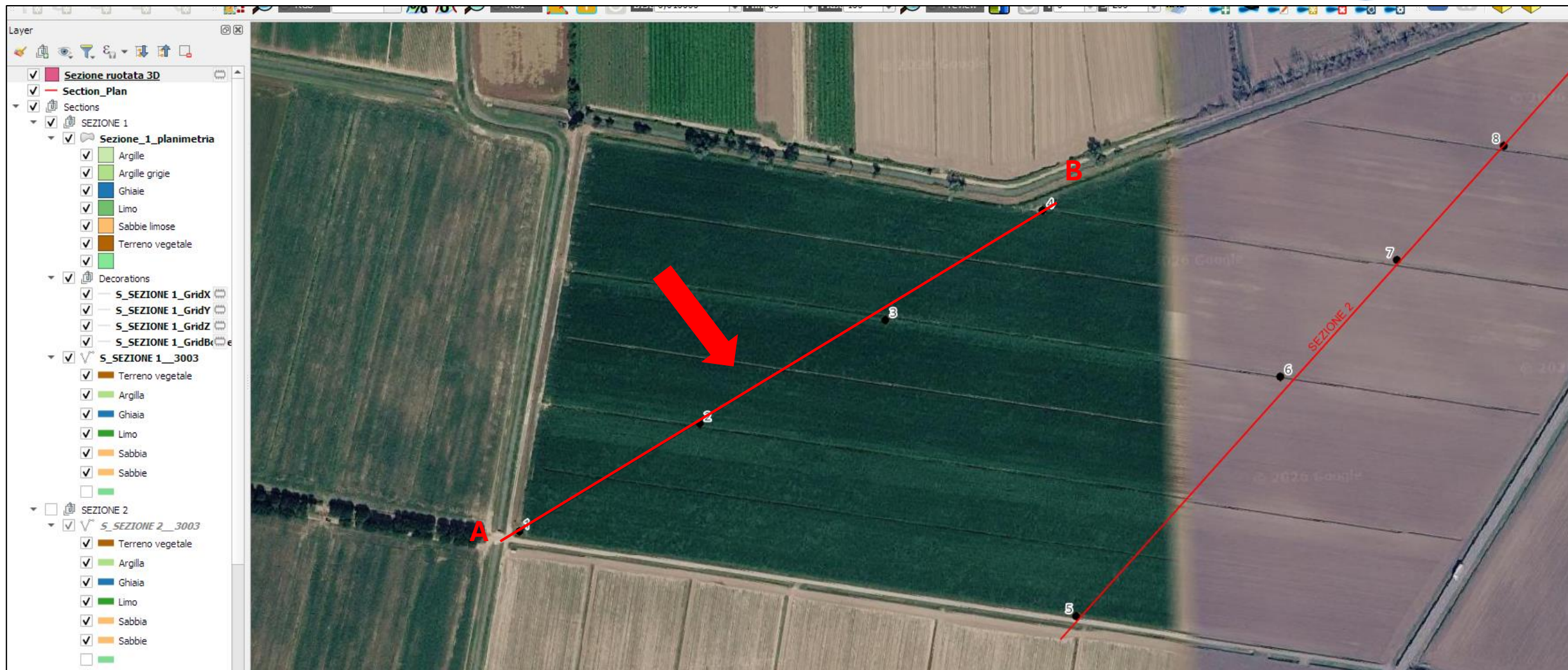
Posiziona e ruota una sezione 2D nello spazio 3D reale usando due punti: A (estremo sinistro) e B (estremo destro). Azimuth e scala orizzontale sono calcolati automaticamente dalla posizione di A e B. L'angolo di rotazione inclina la sezione attorno alla linea A-B (0° = verticale come disegnata, 90° = sdraiata). L'esagerazione verticale moltiplica la quota locale prima della rotazione (x2, x2.5, x5, x10 o un valore personalizzato). L'offset Z sposta l'intera sezione su o giù dopo la rotazione.

- A – Select the shapefile of the interpreted section
- B – Select the rotation axis
- C,D – On the map canvas, pick the start point (A) and end point (B) of the section trace in plan view
- E – Set the rotation angle in degrees
- F – Set the vertical exaggeration to apply to the section; you can also enter a custom value
- H – Want the section to “float” above the DEM? Set an offset.

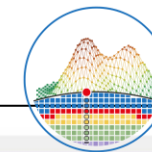
Set the parameters



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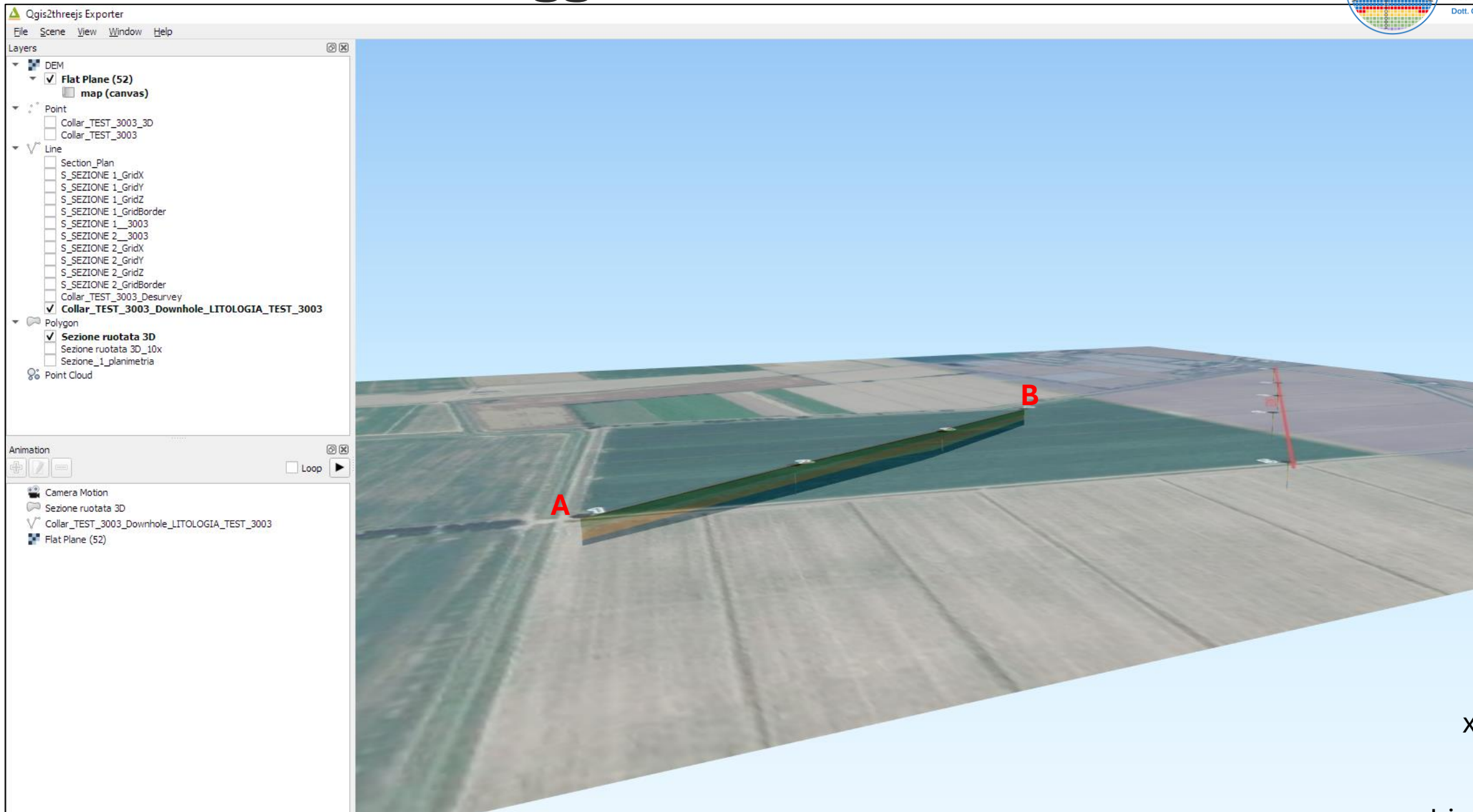


3D view, vertical exaggeration 1x



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3D view, vertical exaggeration 10x

